Embedded Systems Learning Roadmap (Ascending Order of Preparation)

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Who is this for?

- Embedded learners with 1+ year of experience
- Targeting professional skill-building with ESP32, STM32, and sensors

Level 1: Foundations (Beginner)

1. Digital Electronics Fundamentals (Optional Refresh)
   - Logic gates, pull-up/down resistors, voltage levels
2. Microcontroller Basics
   - Microcontroller vs microprocessor
   - CPU, RAM, Flash, I/O ports
   - Example MCUs: AVR, ARM Cortex-M, ESP32, STM32
3. GPIO (General Purpose I/O)
   - Output: LED, buzzer control
   - Input: Push button, switch reading
   - Pull-up/pull-down, active-low/high logic
   - Debouncing methods (manual, software)
4. Timing
   - delay() vs millis() (blocking vs non-blocking)
   - Timer-based LED blinking
5. ADC (Analog-to-Digital Conversion)
   - Using analogRead() to read sensors
   - ADC resolution, voltage reference
   - Filtering techniques (e.g., moving average)
6. PWM (Pulse Width Modulation)
   - LED brightness control
   - Servo motor control
   - Timer-based PWM generation

Level 2: Communication Protocols (Intermediate)

7. Serial (UART)

- Using Serial Monitor: Serial.print(), Serial.read()
- Data logging and command-based control

8. I2C

- Start/stop conditions, addressing, data flow
- Read/write to devices like BNO055

9. SPI

- MOSI/MISO/SCK/CS
- SPI modes (CPOL/CPHA)
- Interfacing with MAX6675, OLED, SD cards

Level 3: Code Architecture & Drivers

10. Modular Embedded C++

- Split code into .h and .cpp
- Write hardware drivers (LED, Button, Sensor)
- Encapsulation and abstraction

Level 4: System-Level Programming

11. Bare-Metal Programming

- No Arduino/OS abstractions
- Access registers directly
- GPIO, ADC, UART using datasheet/register map

12. Interrupts

- External interrupts (GPIO)
- Timer interrupts
- ISR design, NVIC

13. Real-Time Operating System (RTOS)

- FreeRTOS or CMSIS-RTOS
- Task creation, scheduling
- Semaphores, queues, software timers

Level 5: System Integration & Optimization

14. EEPROM & Flash Memory

- Store configuration/settings across resets
- EEPROM access (ESP32 Preferences, STM32 Flash)

15. Low Power Modes

- Sleep, deep sleep, standby
- Wake-up using timer, GPIO, UART

16. Bootloaders & Firmware Updates

- What is a bootloader?
- OTA updates (ESP32)
- STM32 DFU, ST-Link

Level 6: Debugging & Cloud Integration

17. Debugging and Testing

- Serial debugging
- Logic analyzer, oscilloscope (optional)
- JTAG/SWD for step-debugging (STM32, ESP32)
- Unit testing (PlatformIO + Unity)

18. IoT Connectivity

- Wi-Fi, MQTT, HTTP (ESP32)
- Firebase, Blynk, Thingspeak integration
- OTA via web or app

Advanced (Optional)

19. Embedded Linux

- Raspberry Pi, BeagleBone
- Linux kernel basics
- Device tree and Linux driver writing

Final Capstone Projects

- Smart gas and fire alert system with Wi-Fi and OTA
- Wearable fall detector using BNO055
- Cloud-connected datalogger (MQ2 + MAX6675 + ESP32)
- RTOS-based multi-task system with OLED + sensors

Summary (One Glance)

Level	Focus Areas
Beginner	GPIO, ADC, PWM, UART
Intermediate	I2C, SPI, Modular C++, Bare-Metal
Advanced	Interrupts, RTOS, EEPROM, Power Modes
Expert	OTA, Cloud, Debugging, Embedded Linux (opt.)